

1. Abstract

Password security is a crucial aspect of modern digital systems. This project presents a Password Strength Checker developed using C++. The program evaluates the strength of a password based on its length, inclusion of uppercase and lowercase characters, digits, and special symbols. It also checks against commonly used weak passwords. Based on these parameters, a score is calculated and categorized into different levels such as Weak, Moderate, Strong, and Very Strong. This system helps users understand the importance of secure password creation and encourages better cybersecurity practices.

2. Introduction

With the rapid growth of digital platforms, securing user data has become essential. Passwords are the first line of defense against unauthorized access. However, many users still create weak passwords that are easy to guess or hack. This project is implemented using C++ and focuses on analyzing password strength using logical conditions. The program provides feedback to users about the quality of their passwords and promotes awareness about strong password practices.

3. Objectives

1. To design a system that evaluates password strength based on multiple security parameters. 2. To identify weak and commonly used passwords and penalize them. 3. To encourage users to create strong and secure passwords. 4. To implement a simple and efficient password checking mechanism using C++.

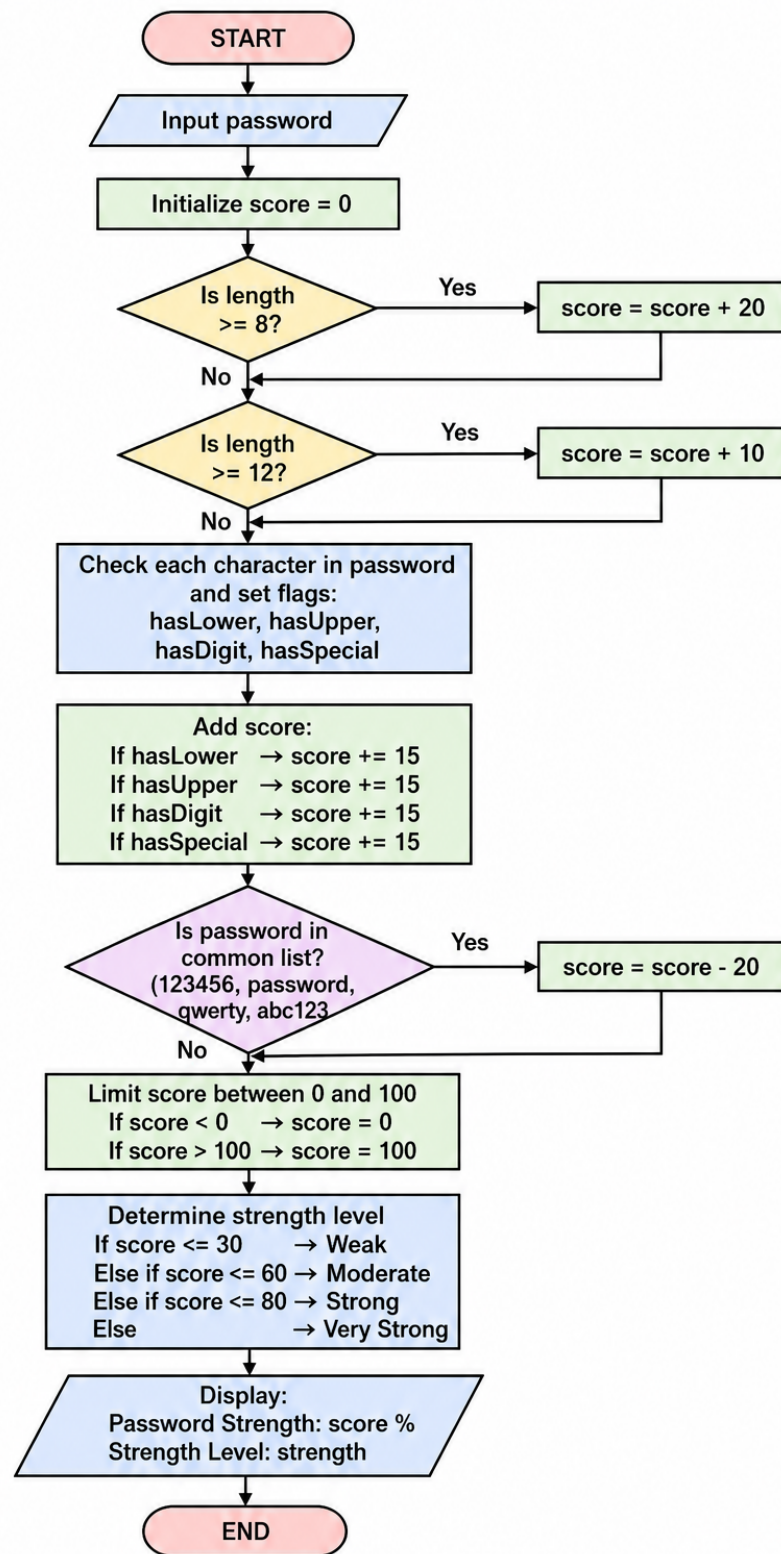
4. Requirements

Software Requirements: - C++ Compiler (GCC, CodeBlocks, Turbo C++) - Operating System: Windows/Linux/Mac Hardware Requirements: - Minimum 4GB RAM - Basic processor (Intel/AMD) - Keyboard and display unit These requirements are minimal as the project is lightweight and console-based.

5. Algorithm

Step 1: Start the program. Step 2: Prompt the user to enter a password. Step 3: Initialize the score variable to 0. Step 4: Check if password length is greater than or equal to 8, add 20 points. Step 5: Check if password length is greater than or equal to 12, add 10 additional points. Step 6: Initialize flags for lowercase, uppercase, digits, and special characters. Step 7: Traverse each character in the password. Step 8: Set corresponding flags if character type is found. Step 9: Add 15 points for each character type present. Step 10: Compare password with common passwords list. Step 11: If match found, subtract 20 points. Step 12: Ensure score remains between 0 and 100. Step 13: Determine strength category based on score. Step 14: Display the password strength and category. Step 15: End the program.

6. Flowchart



7. Methodology

The methodology of this project involves a step-by-step evaluation of password strength. First, the user inputs a password. Then, the program checks for length requirements and different character types. Each satisfied condition contributes to the score. A comparison with commonly used passwords ensures weak patterns are penalized. Finally, the score is categorized into strength levels and displayed to the user. This systematic approach ensures accurate evaluation.

8. Project Implementation

The project is implemented using C++ programming language. It uses standard libraries such as `iostream`, `string`, and `cctype`. The program uses conditional statements, loops, and arrays to evaluate the password. The logic is simple, efficient, and suitable for beginners to understand password validation concepts.

9. Results

The program successfully evaluates different passwords and assigns an appropriate strength score. Weak passwords receive low scores, while strong passwords with multiple character types and sufficient length receive high scores. The output clearly displays both numerical score and descriptive strength level.

10. Future Scope

This project can be enhanced by adding graphical user interfaces (GUI), real-time password suggestions, and integration with login systems. Advanced features such as dictionary attack detection and password history tracking can also be implemented to improve security further.

11. References

1. C++ Programming Language by Bjarne Stroustrup 2. GeeksforGeeks – Password Strength Checker 3. W3Schools C++ Tutorials 4. Cybersecurity Basics Resources 5. Online Programming Documentation